

Sign Selection Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/sign

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Abstract

We study $\text{sign}(x) \in \{-1, 0, 1\}$ under $|x' - x| \leq \varepsilon$. Positive/negative signs are preserved when $|x| > \varepsilon$; the origin is stable only for $\varepsilon = 0$. Lean `LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

1 Operator

Definition 1. $\text{sign}(x) = 1$ if $x > 0$, -1 if $x < 0$, and 0 if $x = 0$.

2 Theorems

Theorem 1 (Preservation). *If $|x' - x| \leq \varepsilon$: (1) $x > \varepsilon \Rightarrow \text{sign}(x') = 1$; (2) $x < -\varepsilon \Rightarrow \text{sign}(x') = -1$; (3) $\varepsilon = 0 \wedge x = 0 \Rightarrow \text{sign}(x') = 0$.*

Theorem 2 (Sharpness). *If $0 < x \leq \varepsilon$ then some admissible x' has $\text{sign}(x') \neq 1$; symmetrically on the negative side; if $x = 0$ and $\varepsilon > 0$ then some x' has $\text{sign}(x') \neq 0$.*

3 Proofs

Preservation follows from $|x' - x| \leq \varepsilon$ and ω -arithmetic. Sharpness uses $x' = x \pm \varepsilon$ or $x' = \varepsilon$ at the origin.

4 Comparison

Relative to Threshold: shared cut geometry at 0, but ternary output and strict nonzero convention.
Relative to Multi-threshold: single cut, richer label set.

5 Lean / certificate

Research.Operators.Sign.Preservation; certificate `lean/certificates/sign/sign-preservation/`
(`LEAN_FULL`; Mathlib $\mathbb{R}$).

6 Limitations

Mathlib $\mathbb{R}$ (`REAL_MATHLIB`); no privacy claims.